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APPLIED RESEARCH

Beyond Random Splits: A Temporal Robustness Benchmark for Industrial Food Anomaly Detection

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ABSTRACT Industrial visual anomaly detection has achieved near-saturated performance on benchmarks such as MVTec AD, yet deployed systems face challenges these benchmarks do not capture: deformable subjects, non-laboratory imaging, and temporal distribution shift. We introduce RoboCook-TAD, a benchmark of 10,124 images collected over six months from the quality-inspection camera of a commercially deployed udon-cooking robot. The dataset contains 4,512 normal frames, 2,083 foreign-object images with 5,219 bounding-box annotations, 1,847 process anomalies, and 1,682 operational anomalies for extended evaluation. RoboCook-TAD adopts a temporal train/test split as its primary protocol and introduces the Temporal Generalization Gap (TGG), defined as the AUROC difference between random and temporal splits for each method. We benchmark twelve methods spanning unsupervised, semi-supervised, supervised, and zero-shot families under a unified protocol. Key findings: 1) state-of-the-art methods drop from 99% AUROC on MVTec AD to 74–86% on our data; 2) method rankings change between protocols, with 2–10 AUROC points attributable to train/test similarity; 3) fully-supervised detectors achieve the highest random-split AUROC but the largest TGG, while semi-supervised methods offer the best trade-off; 4) temporal degradation correlates strongly with measurable feature-space distributional drift; and 5) on an extended open-set evaluation of operational anomalies, zero-shot vision-language models outperform all other families, revealing that visual detection alone is insufficient for complete quality assurance.

INDEX TERMS Anomaly detection, food inspection, robotic cooking, temporal distribution shift, benchmark dataset, foreign object detection.

I. INTRODUCTION

Autonomous robotic systems are rapidly entering commercial food preparation, from beverage dispensing kiosks to full-service noodle and stir-fry robots deployed in airports, cafeterias, and convenience stores. As these systems take over an increasing share of consumer-facing cooking, *visual quality assurance* shifts from a human responsibility to a computer-vision problem: every dish leaving the machine must be verified as free of foreign objects and conforming to the intended preparation. A single missed contaminant (a hair, a fragment of packaging, a stray paper clip) is not merely a quality issue but a food-safety incident with regulatory and reputational consequences. The need for reliable, automatic, and continuously deployable visual anomaly detection in this setting is therefore both urgent and underserved.

Industrial visual anomaly detection has matured rapidly over the past five years, driven largely by the MVTec AD [1], VisA [2], BTAD [3], and MPDD [4] benchmarks and a series of strong unsupervised methods including PatchCore [5], EfficientAD [6], Reverse Distillation [7], FastFlow [8], SimpleNet [9], and CFLOW-AD [10]. On the established benchmarks, several of these methods report image-level

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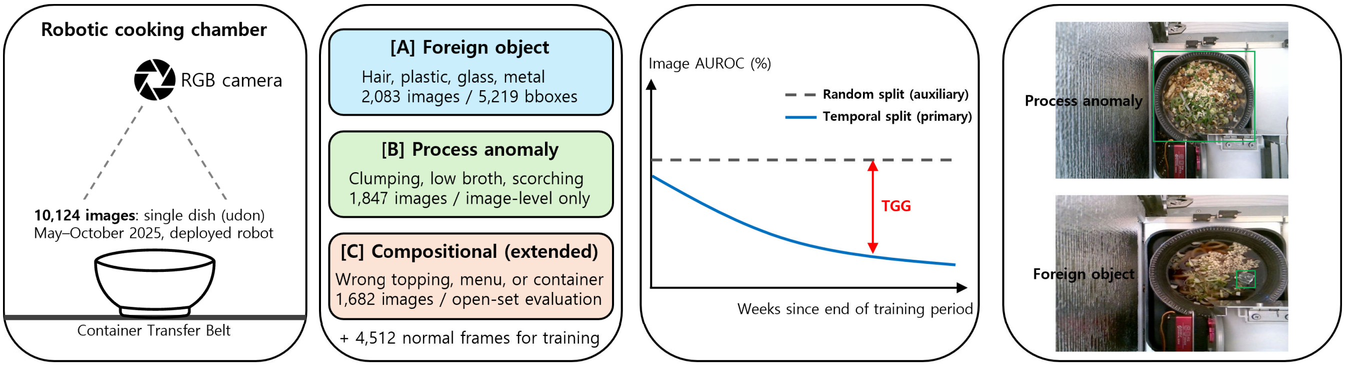

(a) Collection Setup (b) Anomaly Taxonomy (c) Image AUROC vs. Weeks (d) Sample Images

FIGURE 1. Overview of RoboCook-TAD. From left: (i) fixed RGB camera inside a deployed udon-cooking robot, yielding 10,124 images over six months; (ii) three anomaly categories – foreign object (A) and process anomaly (B) form the main benchmark, compositional (C) is reserved for extended open-set evaluation; (iii) near-flat random-split AUROC versus degrading temporal-split AUROC, whose gap we term the *Temporal Generalization Gap*(TGG); (iv) qualitative examples – a process anomaly annotated with a bowl-level box and a foreign-object contamination annotated with a tight box. Curves in panel (iii) are illustrative; quantitative results.

AUROC above 99%, leading to a widespread perception that industrial visual anomaly detection is approaching saturation. We argue that this perception is an artifact of three structural choices shared by virtually all existing benchmarks, each of which is violated in real food-preparation deployments:

1) **Rigid, manufactured inspection subjects.** Existing benchmarks target objects whose normal appearance is essentially invariant up to pose. Food, by contrast, is *deformable*, ingredient-rich, and visually irregular even when nothing is wrong: noodles in random configurations, scattered toppings, steam, and surface reflections make "normal appearance" fundamentally distributional rather than template-like.
2) **Laboratory imaging conditions.** Images in MVTec AD and its successors are captured in dedicated imaging rigs, introducing a laboratory-to-deployment gap that is rarely measured and often substantial.
3) **Randomly partitioned train/test splits.** Every major benchmark partitions data by random sampling from a single short collection session, meaning train and test images share the same illumination, camera state, hardware wear, and ingredient batches. The conventional protocol evaluates interpolation within one session, not generalization to the following weeks or months.

A practical robotic cooking deployment violates all three assumptions simultaneously. The inspection subject is non-rigid food. The camera is the operating quality-control sensor of the robot, not a laboratory rig. And the model, once deployed, must classify dishes for weeks or months under steadily drifting conditions: fluorescent lights age, the gripper accumulates micro-scratches, the foil-lined chamber walls dull, ingredient suppliers change between batches, and seasonal humidity alters how steam settles on the broth. Whether existing anomaly detectors degrade gracefully under such drift – and, if so, by how much – is currently unknown because no public benchmark is structured to ask the question.

A. OUR BENCHMARK

We introduce **RoboCook-TAD**,[1] a benchmark of 10,124 images collected over a six-month period (May–October 2025) from the quality-inspection camera of a commercially deployed udon-cooking robot. Figure 1 summarizes the benchmark. Every image is the exact frame the robot's camera captured as the bowl sat on the container transfer belt immediately before delivery, with no staging or post-hoc curation. The camera position, viewpoint, dish recipe, bowl, and transfer belt are held constant by design; the variability of interest lies in the *anomalies* and in the *passage of operating time*.

As illustrated in Figure 1, the dataset contains 4,512 normal frames and three categories of anomalies. The *foreign object* category (A) comprises 2,083 images with 5,219 bounding boxes spanning four sub-types (linear, colored, transparent, and metallic contaminants). The *process anomaly* category (B) contains 1,847 images of global cooking failures evaluated at image level. The *operational* category (C) contains 1,682 images of semantic or mechanical deviations reserved for extended open-set evaluation in Section VI.

The benchmark is designed around three principles. First, it is collected in the *deployment environment itself*, eliminating the laboratory-to-deployment gap of prior benchmarks. Second, it adopts a temporal train/test split as its primary evaluation protocol: the first 70% of the collection window for training, the next 10% for validation, and the final 20% for testing, with a conventional random split provided only as an auxiliary. Third, it supports three learning paradigms (unsupervised, semi-supervised, and supervised) under a

[1] **T**emporally-stratified **A**nomaly **D**etection in **Robo**tic **Cook**ing.

single unified evaluation protocol, enabling fair cross-family comparison.

B. RESEARCH QUESTIONS

The benchmark is built to answer three research questions, each of which we believe carries implications well beyond food inspection.

- **RQ1.** How do state-of-the-art industrial visual anomaly detection methods, developed and tuned on rigid manufacturing defects, perform when transferred to deformable food in a realistic robotic cooking environment?
- **RQ2.** How much of the apparent performance of these methods is an artifact of the conventional random-split evaluation? Quantitatively, what is the gap between AUROC measured under a random split and AUROC measured under a temporal split on the same data? The third panel of Figure 1 sketches this gap; we quantify it empirically in Section IV.
- **RQ3.** Does increasing the level of supervision (unsupervised → semi-supervised → fully-supervised) improve or harm robustness to temporal distribution shift? Conventional intuition suggests more labels help; we find the answer is more nuanced.

C. CONTRIBUTIONS

Our contributions are fourfold.

1) **A new large-scale benchmark for visual anomaly detection in robotic food preparation.** RoboCook-TAD comprises 10,124 images from a commercially deployed cooking robot, with image-level labels for all frames and bounding-box annotations for 5,219 foreign-object instances. To our knowledge it is the first anomaly detection benchmark in which data are drawn from an operating quality-control camera rather than a laboratory imaging rig.
2) **A temporal evaluation protocol that exposes performance hidden by random splits.** We adopt a chronological train/test split as the primary protocol and introduce the *Temporal Generalization Gap* (TGG), measuring how much a model's AUROC under a random split exceeds its AUROC under a temporal split. We further partition the test set into weekly buckets, producing the first published temporal-degradation curve for industrial visual anomaly detection.
3) **A unified cross-paradigm evaluation.** We benchmark twelve methods spanning four families: six unsupervised [5], [6], [7], [8], [9], [10], two semi-supervised [11], [12], two fully-supervised [13], [14], and two zero-shot [15], [16]. All are evaluated under identical partitions, conversion pipelines, and metrics, isolating model behavior from protocol confounds.
4) **Empirical findings that challenge prevailing intuition.** State-of-the-art methods that exceed 99% AUROC on MVTec AD drop to 74–86% on RoboCook-TAD, with a further 4–11 point loss under the temporal split. Method rankings change between protocols, and increased supervision does *not* monotonically improve temporal robustness: fully-supervised detectors achieve the highest random-split AUROC but the largest TGG.

II. RELATED WORK

A. INDUSTRIAL ANOMALY DETECTION BENCHMARKS

Modern industrial visual anomaly detection was catalyzed by MVTec AD [1], whose 5,354 images across 15 categories of rigid manufactured objects remain the de facto standard. Its successors expanded along different axes: MVTec LOCO [17] introduced logical anomalies, VisA [2] increased the image count to roughly 10,821, and BTAD [3] and MPDD [4] added further categories. More recent benchmarks have scaled aggressively: Real-IAD [18] collects about 150,000 images across 30 categories from a real production line, and Real-IAD Variety [19] extends this to 198,960 images across 160 categories. VISION Datasets [20] and PKU-GoodsAD [21] further broaden the inspection domains. All of the above share three structural properties violated in our setting: rigid inspection subjects, laboratory or dedicated-rig imaging, and random train/test splits.

Beyond manufacturing inspection, real-world benchmark construction in deployed visual systems has also been studied in surveillance imaging, where Han et al. [22] introduced a CCTV image quality benchmark with entropy-based perceptual analysis, and Han and Kim [23] constructed HIQA-DB for hospital surveillance imaging. Like our setting, these works emphasize that data captured by operating cameras differs systematically from laboratory rigs.

The closest prior work to ours is AeBAD [24], which constructs an aero-engine blade dataset in which the test-set normal distribution exhibits a domain shift induced by illumination and viewpoint changes. AeBAD is the first industrial anomaly detection benchmark to make distribution shift a first-class evaluation axis. Its shift, however, is environmental rather than temporal, it covers a single rigid subject, and its protocol compares matched versus shifted splits rather than a continuous temporal curve. Our benchmark holds the imaging rig fixed and lets the passage of operating time be the sole source of drift.

In the food domain, deep-learning approaches have been dominated by X-ray imaging of packaged products [35], [36]. Choi et al. [37] move to RGB imaging with YOLOv5s for fresh-cut vegetables, noting that X-ray systems achieve only 61.8% detection rate on non-metallic materials [38]. These datasets target packaged food on conveyor belts, are small and single-session, and rely on synthetic anomalies. None target cooked food from an autonomous robot, and none employ a temporal evaluation protocol. RoboCook-TAD is, to our knowledge, the first visual anomaly detection benchmark for food that uses RGB imaging from a deployed cooking robot, contains real anomalies, and evaluates methods across time.

B. METHODS FOR VISUAL ANOMALY DETECTION

Visual anomaly detection methods group naturally by the amount of supervision they require. *Unsupervised* methods learn a model of normal appearance from defect-free training images and score test images by their deviation from that model. Three sub-families dominate: memory-bank methods such as PaDiM [25] and PatchCore [5]; student–teacher distillation methods including Reverse Distillation [7] and EfficientAD [6]; and normalizing-flow methods such as FastFlow [8] and CFLOW-AD [10]. SimpleNet [9] bridges these families with a feature adapter, while DRAEM [26] and CutPaste [27] convert anomaly detection into supervised segmentation on synthetic anomalies.

Semi-supervised methods exploit a small number of real anomaly samples. DevNet [11] trains an anomaly scorer against a reference distribution, DRA [28] and PRN [29] model the heterogeneity of seen versus unseen anomalies, and BGAD [12] introduces an explicit separating boundary from a normalizing flow. *Vision-language* methods represent the newest line: WinCLIP [15] adapts CLIP through compositional prompts, and AnomalyCLIP [16] learns object-agnostic prompts focusing on normality. We evaluate representative methods from each regime; to our knowledge, no prior benchmark has compared all three under a single unified protocol.

C. ANOMALY DETECTION UNDER DISTRIBUTION SHIFT

A growing body of work recognizes that the near-saturated performance reported on MVTec AD is partially an artifact of the match between train and test distributions within a single imaging session. Cao et al. [30] formulate anomaly detection under distribution shift as an out-of-distribution generalization problem and show that standard detectors degrade substantially when illumination or background change. A recent empirical study [31] further finds that model rankings on perturbed data differ markedly from those on clean data. RoDA [32] proposes domain adaptation over a normal memory bank using optimal transport. In the surveillance video domain, Kwon and Kim [33] proposed an online continual learning architecture for intrusion detection that explicitly addresses the non-stationarity of long-running deployments, and Song et al. [34] apply continual learning to personalized abnormal behavior recognition. These works share our motivation that deployed visual recognizers must adapt to drifting distributions, but address surveillance video rather than industrial inspection and lack a temporal evaluation protocol. The community increasingly recognizes that distribution shift is the dominant obstacle to deploying anomaly detectors in production, yet no public benchmark makes *temporal* shift the primary evaluation axis. RoboCook-TAD fills that gap.

III. THE ROBOCOOK-TAD BENCHMARK

A. COLLECTION SETUP

All images in RoboCook-TAD are captured by a fixed RGB camera mounted inside the cooking chamber of a commercially deployed udon-cooking robot, looking down at the serving bowl as it sits on the container transfer belt immediately before delivery to the customer. The camera resolution, optical configuration, illumination, container, and dish recipe are held constant across the entire collection: every frame depicts the same menu item in the same bowl on the same belt, photographed from the same viewpoint. This deliberate constancy is a defining property of the benchmark, not an oversight, and is justified in Section III-H.

Data collection spans six months of normal commercial operation, from May to October 2025. No staging, restaging, or post-hoc photographic curation was performed: every frame is exactly what the robot's quality-inspection camera observed at delivery time during its actual service to paying customers. This eliminates the laboratory-to-deployment gap that limits prior benchmarks. Figure 2 shows representative frames for each category. The visible variation among normal samples in the first column, including noodle configuration, topping distribution, broth reflections, and steam, is intrinsic to the cooking process and constitutes the ''normal appearance'' that all evaluated methods must learn to model.

B. ANOMALY TAXONOMY

Anomalies in RoboCook-TAD are organized into three top-level categories (Figure 2, columns 2–4). Categories A (foreign object) and B (process anomaly) constitute the *main benchmark* evaluated throughout Sections IV and V. Category C (operational anomaly) is reserved for extended evaluation in Section VI. This separation is intentional: A and B share a unified evaluation protocol grounded in visual anomaly detection, while C probes system-level failure modes that require fundamentally different reasoning.

Category A – Foreign object. Discrete physical contaminants including hair, thin wire, colored plastic or vinyl fragments, transparent glass or plastic shards, and metallic objects such as paper clips (Figure 2, column 2). Each instance is annotated with a tight axis-aligned bounding box; a single image may contain multiple instances. The objects range from sub-millimeter-thick hair barely distinguishable from green-onion edges to centimeter-scale metallic clips, creating a wide difficulty spectrum within one category.

Category B – Process anomaly. Failures of the cooking process in which the dish is not in a deliverable state despite no foreign contaminant being present (Figure 2, column 3). Observed modes include absence of noodles, noodle clumping, ingredient spillage, and insufficient portion. These anomalies are inherently global properties of the dish; they are annotated with a bowl-encompassing box that serves only as an image-level positivity marker and is never scored under localization metrics.

Category C – Operational anomaly (extended evaluation only). Cases in which the robot system itself malfunctions, rather than the food exhibiting a visual defect (Figure 2, column 4). This category encompasses two families of failure: *semantic mismatch*, in which the delivered dish does

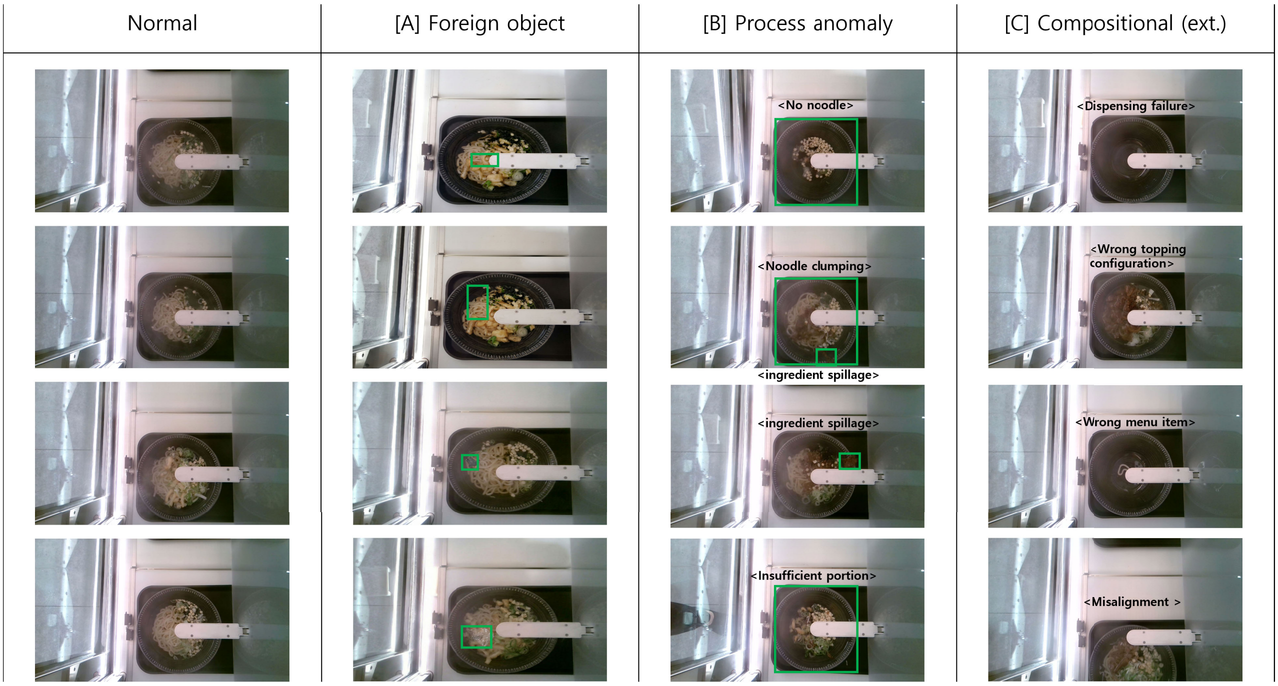

FIGURE 2. **Representative samples from RoboCook-TAD arranged by category. *Column 1*: normal frames exhibiting the natural variability of a correctly prepared udon bowl. *Column 2*: foreign-object contamination (Category A), each annotated with a tight bounding box around the contaminant. *Column 3*: process anomalies (Category B) – no noodle, noodle clumping, ingredient spillage, and insufficient portion, each annotated with a bowl-encompassing box. *Column 4*: operational anomalies (Category C) – dispensing failure, wrong topping configuration, wrong menu item, and misalignment.**

not match the intended order (wrong topping configuration, wrong menu item), and *mechanical or positional failure*, in which the physical delivery process goes wrong (dispensing failure yielding an empty or near-empty bowl, and bowl–camera misalignment in which the bowl is displaced from the camera's field of view). These cases are visually diverse and often indistinguishable from normal frames by pixel-level deviation alone, requiring open-set or contextual reasoning that conventional anomaly detectors do not provide.

C. WHY BOUNDING BOXES RATHER THAN PIXEL MASKS

We annotate Category A foreign objects with tight axis-aligned bounding boxes rather than pixel-level masks. Since pixel masks are the de facto standard in industrial anomaly detection benchmarks [1], [2], [18], this choice requires explicit justification. We argue that bounding boxes are the *appropriate* granularity for our setting for three reasons.

First, foreign-object contamination is object-centric, not region-centric. The defects in MVTec AD (cracks, scratches, color blotches) are naturally connected pixel regions whose spatial extent is part of their semantic content. Foreign objects have no such property: a hair is one hair regardless of pixel footprint, and a paper clip is one paper clip regardless of orientation. The downstream food-safety action (reject the dish, alert the operator) depends on presence and approximate location, not on pixel-precise outlines. This is exactly the abstraction that COCO [39] and LVIS [40] adopt for everyday objects, and we adopt it for the same reason.

Second, pixel-level annotation is unreliable for the dominant object sizes in our data. As Table 1 shows, the median bounding-box area across all foreign-object instances is only 198 pixels, and for the linear sub-type (hair, wire) it drops to 46 pixels. Objects of this size are one to three pixels wide; a pixel mask is dominated by sub-pixel quantization noise, and two annotators marking the same hair will disagree on the majority of boundary pixels. We verify this empirically in Section III-D: pixel-level inter-annotator IoU on an 80-instance feasibility study is 0.34 overall and 0.19 on linear objects, far below the 0.7–0.8 range typical of rigid defects in MVTec AD. Any pixel AUROC computed against such masks would measure annotation noise rather than model performance.

Third, pixel-perfect localization carries no decision value in this domain. A contaminated dish is rejected wholesale; the precise outline of the contaminant is never consumed by the downstream pipeline. Bounding-box localization is sufficient for every operationally meaningful decision and supports the standard COCO-style evaluation toolchain.

D. ANNOTATION PROTOCOL AND INTER-ANNOTATOR AGREEMENT

All images carry an image-level category label (`normal`, `A`, `B`, or `C`). Category A images additionally carry a tight bounding box per foreign-object instance; Category B images carry a bowl-encompassing box as a positivity marker;

TABLE 1. Bounding-box size distribution of Category A foreign-object instances at native camera resolution (1920 × 1080; image area ≈ 2.07 × 10^6 px).

Object type	# inst.	Mean (px^2)	Median (px^2)	Rel. area (%)
Linear (hair, wire)	1,901	82	46	0.002
Colored fragment	1,398	421	283	0.014
Transparent fragment	819	312	198	0.010
Metallic	1,101	953	724	0.035
All foreign obj.	*5,219*	*407*	*198*	*0.010*

TABLE 2. Composition of RoboCook-TAD under the temporal evaluation protocol. Categories A and B form the main benchmark; Category C is reserved for the extended evaluation in Section VI. The # bbox column counts individual foreign-object instances (Category A only; a single image may contain multiple instances).

Category	Train	Val	Test	Total	# bbox
Normal	3,158	451	903	4,512	—
A – Foreign obj.	1,458	208	417	2,083	5,219
B – Process	1,293	185	369	1,847	—
Main (Norm. + A + B)	5,909	844	1,689	8,442	5,219
C – Operational	1,682 (extended eval. only)				—
Total	—	—	—	*10,124*	*5,219*

Category C images carry an image-level label only. Labels were produced by three trained annotators following a written guideline finalized prior to annotation.

To quantify label reliability, we conducted an inter-annotator agreement audit on a uniformly sampled 8% subset, independently re-annotated by a second annotator. Four-way category agreement reaches Cohen's $\kappa = 0.87$ ("almost perfect" on the Landis–Koch scale), and binary foreign-object agreement is $\kappa = 0.93$. Mean bounding-box IoU on Category A instances is 0.71 (median 0.78), highest on metallic objects (0.82) and lowest on linear objects (0.58), consistent with the difficulty gradient induced by object size.

To empirically support the annotation choice of Section III-C, the same two annotators produced pixel masks for a stratified sample of 80 Category A instances. Mean pixel IoU is 0.34 overall, dropping to 0.19 on linear objects and 0.22 on transparent fragments, well below the 0.7–0.8 range typical of MVTec AD. This confirms that pixel annotation is not viable for this data and that bounding-box annotation is the appropriate representation.

E. DATASET STATISTICS

Table 2 reports the composition of RoboCook-TAD under the temporal evaluation protocol of Section III-F. The main benchmark (Normal+A+B) totals 8,442 images: 4,512 normal frames, 2,083 foreign-object images, and 1,847 process-anomaly images. The foreign-object category contains 5,219 individual bounding-box instances, an average of 2.5 per anomalous image, reflecting the fact that contamination events frequently involve multiple small objects. Category C (1,682 images) is held out from the main benchmark and used only in the extended evaluation of Section VI.

Table 1 characterizes the size distribution of foreign-object instances. The dataset is dominated by extremely small objects: the overall median bounding-box area is only 198 pixels (0.010% of the image), and for linear objects (hair, wire) the median drops to 46 pixels (0.002%). Even the largest object type (metallic) has a median area of 724 pixels, below the COCO small-object threshold of $32^2 = 1{,}024$ pixels. The entire foreign-object category therefore lies within the small-object regime, a regime in which standard IoU $\geq$ 0.5 thresholds penalize essentially correct predictions for sub-pixel localization errors. We respond to this in two ways: by adopting bounding-box annotation rather than pixel masks (Section III-C), and by reporting hit rate at IoU $\geq$ 0.1 alongside standard mAP (Section III-G).

The temporal collection is approximately uniform across the six-month window, with weekly normal-frame counts varying by less than a factor of two and weekly anomaly counts by approximately a factor of three, consistent with the natural variability of contamination events in a real production setting. Per-week counts are provided in the supplementary material.

F. EVALUATION PROTOCOL: TEMPORAL PRIMARY, RANDOM AUXILIARY

Existing benchmarks [1], [2], [18] partition data by random sampling from a single collection session, which overstates model performance when the test distribution drifts over time. We therefore adopt a *temporal* split as the primary protocol and provide a random split only as an auxiliary for cross-benchmark comparability.

The collection window is partitioned chronologically: the first 70% (May 1 – August 22, 2025) for training, the next 10% (August 23 – September 8) for validation, and the final 20% (September 9 – October 31) for testing. The test set is further subdivided into weekly *temporal buckets*, producing the degradation curves in Section V. The auxiliary random split mirrors the same 70/10/20% ratios, sampled at order level to guard against data leakage from near-duplicate frames. For every evaluated model M, we report the Temporal Generalization Gap:

$$\mathrm{TGG}(M) = \mathrm{AUROC}_{\mathrm{random}}(M) - \mathrm{AUROC}_{\mathrm{temporal}}(M). \quad (1)$$

A model with TGG ≈ 0 is robust to temporal drift; a large positive TGG indicates that apparent performance is partly an artifact of train/test similarity. As shown in Section V, TGG ranks methods very differently from AUROC alone.

G. EVALUATION METRICS

Foreign-object anomalies (A) and process anomalies (B) are evaluated with complementary metric suites. At image level, both categories are scored by AUROC, AUPR,

TABLE 3. Image-level anomaly detection results under the temporal evaluation protocol. Category A (foreign object) and Category B (process anomaly) are scored independently; "All" is the micro-average over both categories. Best result per column is bold; best within each family is underlined. All values are percentages; ± indicates standard deviation over three runs.

		Category A (Foreign obj.)				Category B (Process)				All (A + B)			
Family	Method	AUROC	AUPR	F1	BAcc	AUROC	AUPR	F1	BAcc	AUROC	AUPR	F1	BAcc
Unsupervised	PatchCore	78.4	72.1	68.3	71.5	84.7	79.3	75.6	78.2	80.9	75.0	71.4	74.3
	EfficientAD	76.8	70.5	66.7	70.1	83.2	77.8	74.1	76.9	79.4	73.6	69.8	73.0
	Rev. Distill.	73.6	67.2	63.8	67.4	80.1	74.5	70.9	73.6	76.3	70.3	66.8	70.0
	FastFlow	71.9	65.4	61.7	65.8	78.6	72.9	69.2	71.8	74.8	68.6	64.9	68.3
	SimpleNet	74.5	68.1	64.5	68.2	81.3	75.6	71.8	74.5	77.4	71.3	67.6	70.9
	CFLOW-AD	72.8	66.3	62.7	66.6	79.4	73.7	70.1	72.7	75.6	69.4	65.9	69.2
Semi	DevNet	80.9	74.6	70.8	73.9	87.2	82.1	78.3	80.7	83.5	77.8	74.0	76.8
	BGAD	82.3	76.2	72.4	75.4	88.6	83.7	79.8	82.1	84.9	79.3	75.5	78.3
Sup.	YOLOv10	79.5	73.8	69.6	72.7	85.3	80.1	76.2	78.8	81.9	76.4	72.4	75.3
	RT-DETR	80.2	74.5	70.3	73.4	86.1	80.9	77.0	79.6	82.6	77.1	73.1	76.0
ZS	WinCLIP	65.8	59.2	55.7	59.4	73.4	67.1	63.5	66.2	69.1	62.6	59.1	62.4
	AnomalyCLIP	68.3	61.7	58.1	61.8	76.2	69.8	66.1	68.7	71.7	65.2	61.6	64.8
Best overall		**82.3**	**76.2**	**72.4**	**75.4**	**88.6**	**83.7**	**79.8**	**82.1**	**84.9**	**79.3**	**75.5**	**78.3**

F1-max, and balanced accuracy. For localization (Category A only), we report mAP@0.5 and mAP@[.5:.95] following the COCO convention, recall at 1 FPPI, and hit rate at IoU $\geq$ 0.1, an intentionally permissive criterion motivated by the small-object statistics of Table 1. Process anomalies are evaluated at image level only; their bowl-encompassing boxes serve as positivity markers and are excluded from localization scoring.

Unsupervised baselines output per-pixel anomaly heatmaps rather than bounding boxes. To evaluate them under localization metrics, we apply a unified post-processing pipeline: per-image min-max normalization, thresholding at the F1-maximizing value on the validation set, connected-component extraction, and axis-aligned bounding-box emission. The same pipeline and hyperparameters are applied to every method; details are in the appendix.

H. ON THE SINGLE-ITEM, SINGLE-ENVIRONMENT SCOPE

The most likely critique of this benchmark is that it depicts a single dish in a single environment. We argue this scope is a deliberate design principle. Every category in MVTec AD, VisA, BTAD, and MPDD is itself a single object in a single rig under fixed lighting; per-category, RoboCook-TAD contains 10,124 images, more than an order of magnitude larger than any single MVTec AD category and more than double the entire AeBAD dataset [24]. Every frame comes from the operating quality-control camera of a commercial robot, eliminating the train-to-deployment gap entirely. Most importantly, holding menu, container, geometry, and viewpoint fixed isolates temporal distribution shift as the dominant source of variation, making the temporal protocol a clean experimental instrument; a multi-dish dataset would confound temporal effects with category effects.

We frame RoboCook-TAD as the first installment of what we hope will become a multi-dish family, with subsequent releases extending to additional menu items under the same evaluation protocol. The protocol itself is dish-agnostic and is the more durable contribution.

IV. EXPERIMENTS

A. EXPERIMENTAL SETUP

We evaluate six unsupervised methods trained on normal images only (PatchCore [5], EfficientAD [6], Reverse Distillation [7], FastFlow [8], SimpleNet [9], CFLOW-AD [10]), two semi-supervised methods that additionally receive 10 labeled anomalies per sub-type (DevNet [11], BGAD [12]), two fully-supervised object detectors trained on all bounding-box annotations (YOLOv10 [13], RT-DETR [14]), and two zero-shot vision-language models with no target-domain data (WinCLIP [15], AnomalyCLIP [16]).

All unsupervised and semi-supervised methods are implemented within the Anomalib framework using a WideResNet-50-2 backbone pretrained on ImageNet at 256×256 resolution. PatchCore uses a coreset sampling ratio of 1%; EfficientAD uses the small variant (EfficientAD-S). YOLOv10-M and RT-DETR-L are fine-tuned from COCO-pretrained weights for 100 epochs with an initial learning rate of 0.01 and cosine annealing at 640×640 resolution. WinCLIP and AnomalyCLIP use the ViT-L/14@336 CLIP backbone with default prompt templates. All experiments are run on a single NVIDIA A100 (80 GB) GPU; results are averaged over three runs with different random seeds. For unsupervised and semi-supervised methods, we apply the unified heatmap-to-bounding-box conversion pipeline described in Section III-G.

B. MAIN RESULTS UNDER TEMPORAL PROTOCOL

Tables 3 and 4 report image-level and localization results under the primary temporal evaluation protocol. Three observations stand out from the image-level results.

TABLE 4. Localization results on Category A (foreign object) under the temporal protocol. mAP follows the COCO convention; R@1FPPI is recall at one false positive per image; Hit is hit rate at IoU ≥ 0.1.

Family	Method	mAP@.5	mAP@[.5:.95]	R@1FPPI	Hit
Unsup.	PatchCore	18.7	7.2	31.4	52.8
	EfficientAD	16.3	6.1	28.7	49.5
	Rev. Distill.	12.8	4.5	23.1	43.2
	FastFlow	10.4	3.6	19.8	38.6
	SimpleNet	14.1	5.2	25.6	45.9
	CFLOW-AD	11.5	4.0	21.3	40.7
Semi	DevNet	21.4	8.6	35.2	57.3
	BGAD	23.8	9.7	37.9	60.1
Sup.	YOLOv10	34.6	15.3	48.2	71.5
	RT-DETR	**37.2**	**16.8**	**51.6**	**74.3**
ZS	WinCLIP	5.1	1.4	10.3	22.7
	AnomalyCLIP	7.3	2.1	13.8	27.4

First, every method performs substantially worse than on MVTec AD, where the same models routinely exceed 99% AUROC. The best method, BGAD, reaches 84.9% AUROC on A+B, a drop of roughly 15 points, confirming that deformable food under realistic cooking conditions is fundamentally harder than rigid parts in a laboratory rig (RQ1). Second, Category B is consistently 4–8 AUROC points easier than Category A across all methods, as process anomalies alter global dish appearance while foreign objects are small, localized, and often camouflaged. Third, semi-supervised methods outperform all unsupervised methods by 3–5 points on both categories, confirming that even 10 labeled anomalies per sub-type provide a meaningful advantage.

For localization, the small-object nature of the data has a dramatic effect. Even the best overall method (RT-DETR) reaches only 37.2% mAP@0.5, yet the gap to hit rate at IoU ≥ 0.1 is large across all methods (e.g., PatchCore: 18.7% vs. 52.8%), confirming that many predictions are essentially correct but penalized by strict IoU on sub-50-pixel objects. Fully-supervised detectors dominate localization by a wide margin, as expected given their direct access to bounding-box labels.

C. RANDOM VERSUS TEMPORAL: THE GENERALIZATION GAP

Table 5 reports the Temporal Generalization Gap (TGG, Equation 1) for all twelve methods, and Figure 3 visualizes the relationship between random-split and temporal-split AUROC as a scatter plot. Two findings are central to RQ2.

Finding 1: TGG is substantial and method-dependent. TGG ranges from 1.8 (AnomalyCLIP) to 9.9 (RT-DETR), meaning that depending on the method, 2–10 AUROC points of apparent performance evaporate once the evaluation protocol respects the passage of time. The mean TGG across all twelve methods is 5.2 points, which exceeds the gap between many competing methods on MVTec AD – in other words, the performance difference attributed to algorithmic improvement on conventional benchmarks is *smaller* than the artifact introduced by random splitting on our data.

Finding 2: rankings change. Under the random protocol, RT-DETR (92.5%) is the top-performing method, followed by YOLOv10 (91.2%) and BGAD (90.4%). Under the temporal protocol, however, BGAD (84.9%) overtakes both supervised detectors, and DevNet (83.5%) ranks third. RT-DETR and YOLOv10 each drop four positions. The rank reversal between supervised detectors and semi-supervised methods is the most striking result of this table: methods that appear strongest under the conventional protocol are the most sensitive to temporal shift. We analyze the mechanism behind this reversal in Section V.

TABLE 5. Temporal Generalization Gap (TGG). Methods are sorted by TGG in descending order. AUROC$_R$ and AUROC$_T$ denote image-level AUROC (A+B, %) under the random and temporal protocols respectively.

Family	Method	AUROC$_R$	AUROC$_T$	TGG	ΔRank
Sup.	RT-DETR	92.5	82.6	9.9	+4
Sup.	YOLOv10	91.2	81.9	9.3	+4
Semi	BGAD	90.4	84.9	5.5	−2
Semi	DevNet	88.9	83.5	5.4	−1
Unsup.	PatchCore	85.6	80.9	4.7	0
Unsup.	Rev. Distill.	81.7	76.3	5.4	0
Unsup.	SimpleNet	82.5	77.4	5.1	0
Unsup.	EfficientAD	83.8	79.4	4.4	+1
Unsup.	CFLOW-AD	80.1	75.6	4.5	0
Unsup.	FastFlow	79.4	74.8	4.6	0
ZS	AnomalyCLIP	73.5	71.7	1.8	0
ZS	WinCLIP	71.0	69.1	1.9	0

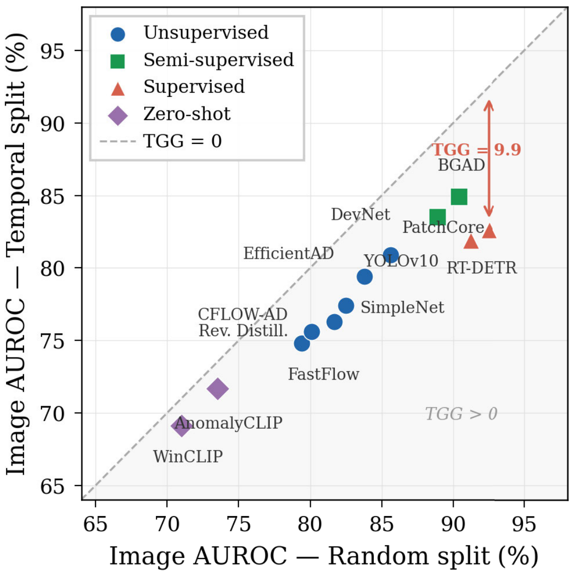

FIGURE 3. Random-split versus temporal-split image AUROC for all twelve methods. The diagonal represents TGG = 0 (perfect temporal robustness). Distance below the diagonal is proportional to TGG.

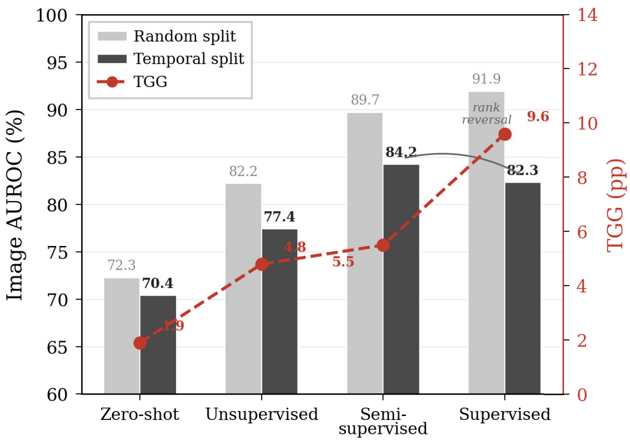

FIGURE 4. Family-level comparison. Left axis: mean image AUROC under the temporal protocol (bars). Right axis: mean TGG (markers). Supervised detectors achieve the highest random-split AUROC but suffer the largest TGG, ending up below semi-supervised methods under the temporal protocol.

D. CROSS-PARADIGM COMPARISON

RQ3 asks whether increasing supervision improves temporal robustness. Table 5 already reveals the answer, but Figure 4 makes the pattern explicit by aggregating metrics at the family level.

Unsupervised methods form the middle ground: moderate absolute AUROC (74.8–80.9% temporal) and moderate TGG (4.4–5.4 points). Within this family, PatchCore and EfficientAD are the most robust; FastFlow and CFLOW-AD lag behind, consistent with the observation that flow-based likelihoods are more sensitive to distributional shift than distance-based memory banks.

Semi-supervised methods (DevNet, BGAD) achieve the best temporal AUROC among all families (83.5–84.9%) with a moderate TGG (5.4–5.5 points). The small set of labeled anomalies helps the model learn a more discriminative boundary without overfitting to the specific appearance of training-period anomalies, because the boundary is anchored by the much larger normal set.

Fully-supervised detectors (YOLOv10, RT-DETR) achieve the highest *random*-split AUROC (91.2–92.5%) but the largest TGG (9.3–9.9 points), ending up at 81.9–82.6% under the temporal protocol – *below* both semi-supervised methods. The mechanism is clear: these models are trained end-to-end on the specific bounding-box annotations of the training period, including the specific appearance, size, and placement of each foreign object. When the visual statistics of contamination events shift even slightly between the training and test periods – a different batch of packaging material, a different hair color, slightly altered broth opacity – the tightly fitted decision boundary fails to generalize. This is the most practically important finding of our evaluation: *the learning paradigm that appears strongest under the conventional random-split protocol is the least robust under realistic temporal deployment conditions.*

TABLE 6. Cross-benchmark performance comparison. MVTec AD AUROC values are taken from the respective original papers or from Anomalib re-implementations under the same backbone. RoboCook-TAD values are from our experiments (image-level AUROC on A+B). The drop column shows the absolute difference between MVTec AD and RoboCook-TAD temporal AUROC.

Method	MVTec	Robo.$_R$	Robo.$_T$	Drop
PatchCore	99.1	85.6	80.9	18.2
EfficientAD	99.1	83.8	79.4	19.7
Rev. Distill.	98.5	81.7	76.3	22.2
FastFlow	97.8	79.4	74.8	23.0
SimpleNet	98.3	82.5	77.4	20.9
CFLOW-AD	98.0	80.1	75.6	22.4
WinCLIP	91.8	71.0	69.1	22.7
AnomalyCLIP	93.2	73.5	71.7	21.5
Mean	*97.0*	*79.7*	*75.6*	*21.3*

Zero-shot methods (WinCLIP, AnomalyCLIP) have the lowest absolute AUROC (69.1–71.7%) but the smallest TGG (1.8–1.9 points), indicating near-perfect temporal robustness. This is expected: these methods never see target-domain data during training, so there is no training-period appearance to overfit to. Their low absolute performance, however, limits their practical utility as standalone detectors. Nevertheless, the near-zero TGG suggests that vision-language features encode a notion of "anomalous" that transfers across time in a way that task-specific features do not – an observation that we believe has implications for future method design.

The overall pattern is *concave*: temporal robustness first improves as supervision increases from zero-shot to semi-supervised (absolute AUROC rises, TGG stays moderate), then *degrades* as supervision increases further to fully-supervised (absolute AUROC peaks on random splits but TGG spikes). This concave relationship between supervision level and temporal robustness is, to our knowledge, reported here for the first time.

E. COMPARISON WITH EXISTING BENCHMARKS

Table 6 compares the eight methods for which MVTec AD results on the same backbone are available. The mean AUROC drop from MVTec AD to RoboCook-TAD under the temporal protocol is 21.3 points, ranging from 18.2 (PatchCore) to 23.0 (FastFlow). Under the random protocol the mean drop is already 17.3 points; the temporal protocol adds a further 4.1 points on average. These numbers quantify two distinct sources of difficulty that our benchmark introduces: the *domain gap* between rigid manufactured objects and deformable food (responsible for the random-split drop) and the *temporal gap* between the training session and the deployment period (responsible for the additional TGG).

One pattern in the cross-benchmark comparison is worth noting: the *relative ranking* of methods is broadly preserved between MVTec AD and the RoboCook-TAD random split, PatchCore and EfficientAD remain the strongest unsupervised methods, but is disrupted under the temporal split,

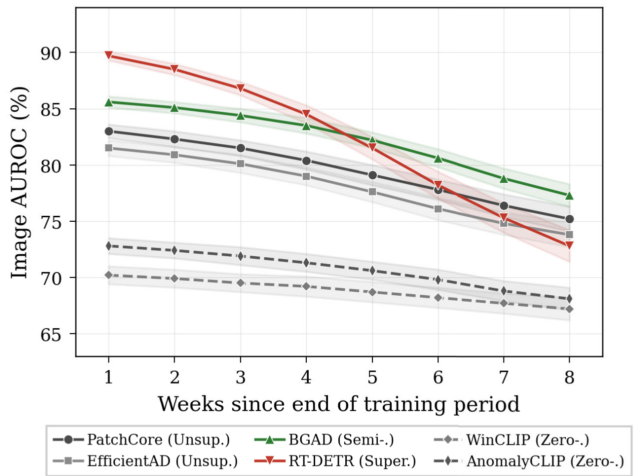

FIGURE 5. Image AUROC (A+B) as a function of weeks elapsed since the end of the training period, for six representative methods. Shaded bands indicate ±1 standard deviation over three runs.

where EfficientAD overtakes PatchCore on Category A. This suggests that the temporal protocol exposes method-specific fragilities that are invisible under conventional evaluation, and that the domain-gap and temporal-gap components of the difficulty stress different aspects of the models.

V. TEMPORAL DEGRADATION ANALYSIS

Section IV established *what* happens under temporal evaluation. This section investigates *why*.

A. PER-BUCKET PERFORMANCE CURVES

Figure 5 plots image-level AUROC (A+B combined) within each weekly temporal bucket for six representative methods, one per sub-family for unsupervised (PatchCore, EfficientAD), one semi-supervised (BGAD), one supervised (RT-DETR), and two zero-shot (WinCLIP, AnomalyCLIP). The test period spans eight weeks (September 9 – October 31, 2025), yielding eight buckets. Five patterns emerge from the curves. First, every method's AUROC decreases monotonically or near-monotonically from bucket 1 to bucket 8, with the mean across all twelve methods dropping from 83.1% to 72.4% over eight weeks. The shift is a property of the data rather than an artifact of any particular model family.

Second, the degradation is non-linear. For unsupervised and semi-supervised methods, weeks one through four show a gentle slope ($\sim$0.5–1.0 points per week), followed by a steeper decline in weeks five through eight ($\sim$1.5–2.5 points per week). This shape suggests that the distribution initially drifts within the variation the model has absorbed during training, then crosses a threshold beyond which the learned normality no longer covers the incoming data. From a deployment perspective, monthly retraining would capture most of the available performance, while quarterly retraining would enter the steeper regime.

Third, supervised methods degrade fastest. RT-DETR starts at the highest bucket-1 AUROC (89.7%) but falls to 72.8% by bucket 8 ($\Delta = 16.9$), nearly double the drop of BGAD ($\Delta = 8.3$) and triple that of AnomalyCLIP ($\Delta = 4.6$). This is consistent with the TGG analysis: end-to-end training on bounding-box annotations memorizes training-period anomaly statistics that drift faster than the global normal-appearance statistics unsupervised methods rely on.

Fourth, zero-shot methods are near-flat, with only ±2–3 points of variation across the entire test window. Vision-language features encode a time-invariant notion of anomalousness, though at the cost of low absolute performance (67–73%). This flatness suggests that such features could serve as a stable prior to be combined with a time-adaptive component (Section VII).

Finally, PatchCore is the most robust unsupervised method, exhibiting the shallowest slope and the highest bucket-8 AUROC (75.2%). Its distance-based memory bank is inherently more tolerant of distributional shift than the likelihood-based scores of FastFlow and CFLOW-AD, consistent with findings on AeBAD's environmental shift [31] and extending them to the temporal regime.

B. DISTRIBUTIONAL DRIFT AND ITS CORRELATION WITH PERFORMANCE

The temporal degradation curves demonstrate that performance drops over time, but they do not explain *what changes in the data*. To answer this, we measure the distributional distance between the training set and each test bucket at two levels.

At the pixel level, we compute the earth mover's distance (EMD) between per-bucket and training-set grayscale intensity histograms (256 bins). EMD increases approximately linearly from bucket 1 (0.021) to bucket 8 (0.089), driven by two causes: gradual yellowing of the fluorescent lighting inside the cooking chamber, and progressive accumulation of oil residue on the chamber walls altering reflectance patterns. Both are mundane consequences of operating a cooking robot for months, precisely the kind of drift that deployed systems must tolerate but laboratory benchmarks never expose.

At the feature level, we compute the Maximum Mean Discrepancy (MMD, Gaussian kernel) between training-set and per-bucket features extracted from a frozen WideResNet-50-2. MMD increases from 0.013 (bucket 1) to 0.071 (bucket 8), confirming that low-level drift propagates into the feature space that anomaly detectors operate on.

Figure 6 and Table 7 establish the quantitative link between drift and degradation. The Pearson correlation between MMD and per-bucket AUROC drop is strong across all twelve methods (mean $r = 0.90$, all $p < 0.01$). Supervised methods are the most drift-sensitive, with RT-DETR and YOLOv10 exhibiting the highest correlations ($r = 0.97$ and 0.96) and steepest regression slopes, consistent with their tightly fitted decision boundaries. Zero-shot methods are least sensitive ($r = 0.78$ and 0.81) with shallow slopes, though they are not

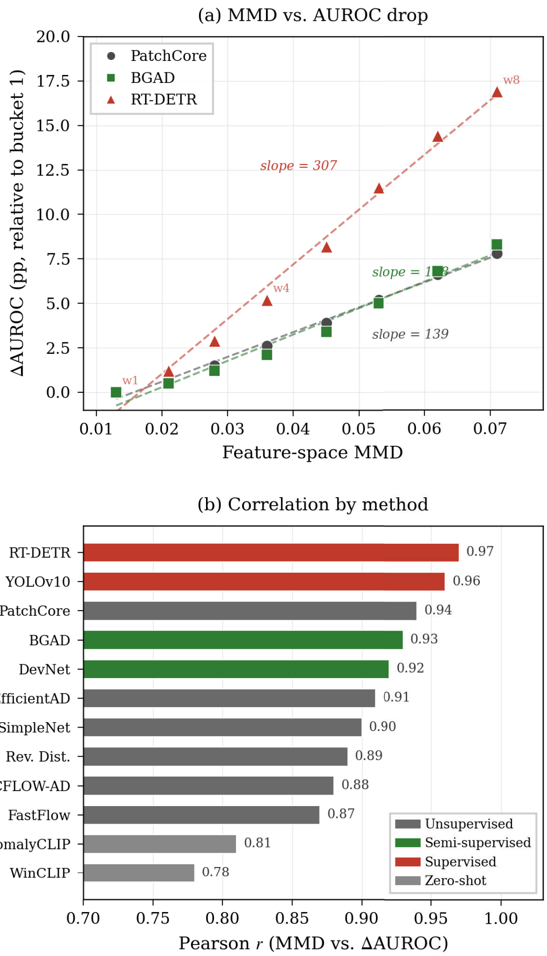

FIGURE 6. **Relationship between distributional drift and performance degradation. (a): feature-space MMD (x-axis) versus image AUROC drop relative to bucket 1 (y-axis) for PatchCore(circles) and RT-DETR (triangles); each point is one weekly bucket. (b): Pearson correlation between MMD and AUROC drop across all twelve methods.**

drift-immune. The approximately linear MMD–ΔAUROC relationship suggests that periodically monitoring MMD could predict when retraining is needed, an observation we formalize in Section VII.

C. QUALITATIVE FAILURE ANALYSIS

To understand *how* models fail as time passes, we manually inspect the false positives and false negatives that appear in buckets 6–8 but not in buckets 1–3. Figure 7 illustrates four recurring failure modes.

(a) Topping false positives. Green-onion fragments that land on the bowl rim or outside the main topping cluster are increasingly flagged as foreign objects in later buckets. This happens because the training-period normal distribution models toppings as a central cluster; as ingredient batch variation shifts the typical topping spread over time, rim-adjacent fragments fall outside the learned normal region. This failure mode accounts for approximately 35% of all false positives in buckets 6–8 and is the single largest source of

TABLE 7. **Pearson correlation (r) between feature-space MMD and per-bucket AUROC drop (ΔAUROC relative to bucket 1) for each method.**

Family	Method	r (MMD vs. ΔAUROC)
Unsup.	PatchCore	0.94
	EfficientAD	0.91
	Rev. Distill.	0.89
	FastFlow	0.87
	SimpleNet	0.90
	CFLOW-AD	0.88
Semi	DevNet	0.92
	BGAD	0.93
Sup.	YOLOv10	0.96
	RT-DETR	0.97
ZS	WinCLIP	0.78
	AnomalyCLIP	0.81
Mean		*0.90*

false alarms for unsupervised methods. It is less prevalent for semi-supervised methods, whose exposure to a few labeled anomalies helps calibrate the boundary between "displaced topping" and "true foreign object."

(b) Reflection false positives. Specular highlights on the broth surface trigger high anomaly scores, particularly in weeks when the chamber lighting has drifted furthest from training conditions. The gradually yellowing fluorescent tubes alter the spectral composition of the highlights, making them appear as novel bright spots to feature-based detectors. This failure mode is especially problematic for normalizing-flow methods (FastFlow, CFLOW-AD), which assign low likelihood to any out-of-distribution feature regardless of its semantic content. Memory-bank methods (PatchCore) are more robust because the distance metric is less sensitive to small spectral shifts than the density estimate.

(c) Missed transparent fragments. Sub-type A3 (transparent glass or plastic) is already the hardest foreign-object type in the early buckets; in late buckets it becomes nearly undetectable. The reason is that the progressive accumulation of oil film on the broth surface reduces the contrast between the transparent contaminant and the liquid, effectively camouflaging the fragment. Detection recall on A3 drops from 38.2% in buckets 1–3 to 19.6% in buckets 6–8 across all methods, a relative decline of 49%. This is the most safety-critical failure mode in our analysis, because transparent foreign objects (broken glass) pose the highest physical risk to consumers.

(d) Missed linear objects. Hair and thin wire (sub-type A1) become harder to detect in late buckets for a related reason: the same oil film that camouflages transparent objects also reduces the contrast of thin dark lines against the broth. Detection recall on A1 drops from 51.7% to 38.4% (relative decline 26%). Unlike transparent fragments, linear objects are partially rescued by the semi-supervised and supervised methods, which have learned shape-based priors

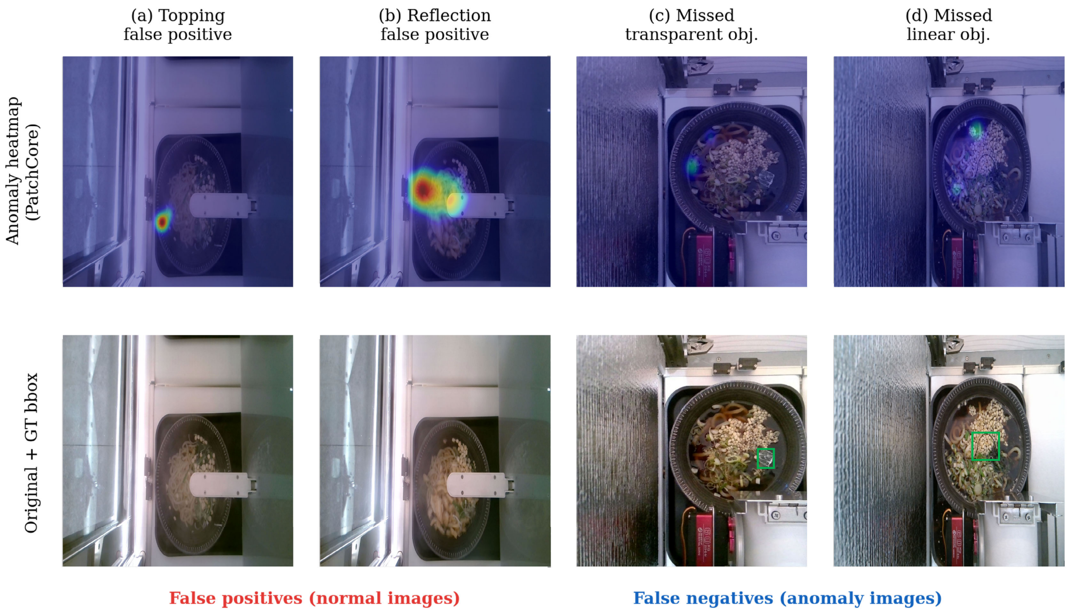

FIGURE 7. Qualitative failure cases that emerge or intensify in late test buckets. Each column shows one failure mode.*Top row*: anomaly heatmaps from PatchCore.*Bottom row*: original images with ground-truth bounding boxes (where applicable). (a) A green-onion fragment near the bowl rim is falsely flagged as a foreign object (FP). (b) A specular reflection on the broth surface triggers a high anomaly score (FP). (c) A transparent fragment (green bbox, bottom) receives no heatmap activation and is missed entirely (FN). (d) A linear foreign object (green bbox, bottom) is camouflaged against the broth and missed (FN). Cases (a)–(b) are normal images with no ground-truth box;cases (c)–(d) are anomaly images whose contaminants the model fails to detect.

for elongated contaminants; RT-DETR's A1 recall drops only from 62.3% to 54.1% (13% relative decline), suggesting that explicit supervision provides some resilience against contrast degradation for objects with distinctive geometric structure.

D. SUMMARY OF TEMPORAL ANALYSIS

The analyses of this section converge on three actionable conclusions: temporal degradation is real, measurable, and predictable ($r = 0.90$); it is non-uniform across both anomaly types and method families; and the underlying mechanisms – environmental aging, contrast loss, and supervision-dependent drift sensitivity – are generic to any long-running visual inspection deployment. The temporal evaluation protocol introduced in this paper is the appropriate tool for surfacing them.

VI. EXTENDED OPEN-SET EVALUATION

The main benchmark (Sections IV–V) evaluates models on Category A and B, which manifest as *visual* deviations from the normal dish. Category C, by contrast, contains *operational* anomalies in which the robot system itself malfunctions, including wrong topping configuration, wrong menu item, wrong container, ingredient spillage, insufficient portion, dispensing failure, and bowl-camera misalignment. Many of these cases are visually similar or identical to normal frames, and detecting them requires contextual or semantic reasoning beyond standard anomaly detection. We therefore evaluate Category C separately as an *extended open-set* challenge.

A. SETUP

We evaluate the same twelve methods from Section IV without any retraining or fine-tuning: models trained on normal images from the main benchmark are applied directly to the 1,682 Category C images. Since Category C carries only image-level labels (no bounding boxes), we report image-level metrics only: AUROC, AUPR, and F1-max. We group the seven Category C sub-types into two families based on the nature of the failure:

- *Semantic mismatch* (SM): wrong topping configuration, wrong menu item, wrong container. These cases contain a correctly cooked dish that does not match the intended order. The visual content is plausible food in a plausible bowl; only comparison with the order specification reveals the error.
- *Mechanical failure* (MF): spillage, insufficient portion, dispensing failure, misalignment. These cases exhibit visually obvious deviations such as an empty bowl, food outside the container, or a displaced bowl, and are detectable without order context but differ qualitatively from the A/B anomalies on which the models were trained.

B. RESULTS

Table 8 reports the results. Three findings stand out. First, semantic mismatch is near-random for feature-based methods. On the SM sub-group, all unsupervised, semi-supervised, and supervised methods achieve AUROC between 51.8% and 56.3%, barely above the 50%

TABLE 8. Image-level results on Category C (operational anomaly) under the temporal protocol. Models are trained on normal images from the main benchmark (A+B) with no exposure to Category C during training. SM = semantic mismatch sub-types; MF = mechanical failure sub-types; All = combined Category C.

		SM			MF			All C		
Family	Method	AUC	APR	F1	AUC	APR	F1	AUC	APR	F1
Uns.	PatchCore	53.2	50.8	50.1	72.6	68.3	64.7	61.4	58.1	56.2
	EfficientAD	52.7	50.4	49.8	71.3	67.1	63.5	60.5	57.3	55.4
	SimpleNet	51.8	49.6	49.1	70.5	66.2	62.8	59.7	56.5	54.7
Se.	DevNet	55.1	52.3	51.5	74.8	70.6	66.9	63.5	60.0	58.0
	BGAD	56.3	53.1	52.3	75.6	71.4	67.8	64.4	60.8	58.8
Su.	YOLOv10	54.8	52.0	51.2	76.2	72.0	68.3	63.9	60.5	58.4
	RT-DETR	55.4	52.6	51.8	77.1	72.8	69.1	64.6	61.2	59.1
ZS	WinCLIP	61.7	57.9	56.2	74.3	69.8	66.1	67.2	63.1	60.5
	AnomalyCLIP	**64.2**	**60.3**	**58.7**	75.1	70.6	67.0	**69.0**	**64.8**	**62.3**

random baseline. These methods learn a distribution over *visual features*, and a wrong-topping or wrong-menu dish is, by visual-feature standards, a perfectly normal-looking bowl. The anomaly is *semantic*, not *visual*, and no amount of patch-level feature comparison can detect it without access to the order context.

Second, mechanical failures are partially detectable. On the MF sub-group, AUROC ranges from 70.5% to 77.1%. Dispensing failure and misalignment are the easiest sub-types, as they produce large-scale visual deviations. Spillage and insufficient portion are harder because the bowl still contains food but in the wrong quantity or distribution. These sub-types sit in the gap between visual anomaly (A/B) and semantic anomaly (SM), serving as a natural stress test for open-set generalization.

Third, zero-shot VLMs are the strongest family on Category C. AnomalyCLIP achieves 69.0% overall AUROC, outperforming all other methods including RT-DETR (64.6%). The advantage is concentrated on the SM sub-group, where AnomalyCLIP reaches 64.2%, 8–12 points above feature-based methods. This is consistent with VLMs' text-aligned representations encoding high-level semantic concepts that are invisible to patch-level extractors. On the MF sub-group the advantage disappears: supervised detectors match or slightly outperform VLMs, since mechanical failures produce visual deviations that bounding-box-trained models handle well.

C. IMPLICATIONS

The Category C results carry a clear practical implication: *visual anomaly detection alone is insufficient for complete quality assurance*. Semantic mismatch is a real and operationally consequential failure mode that no visual-only method we evaluated can detect reliably. A complete food inspection pipeline must combine visual anomaly detection (for A and B) with *order-aware verification* that compares the delivered dish against the intended order specification, whether through a separate classification head conditioned on the order ID, through a vision-language model prompted with the order description, or through a non-visual check (e.g., weight measurement, ingredient tracking from the recipe execution log). The relative strength of VLMs on this task, and the broader design implications for hybrid detection architectures.

VII. DISCUSSION

A. PRACTICAL RECOMMENDATIONS

Our experiments yield three recommendations for practitioners. First, semi-supervised methods should be preferred when labeled anomalies are available. BGAD and DevNet achieve the highest temporal-split AUROC with moderate TGG, demonstrating that even ten labeled anomalies per sub-type provide a substantial advantage without the temporal fragility of fully-supervised detectors. For new deployments where labeled anomalies are initially unavailable, starting with PatchCore and transitioning to a semi-supervised model as labels accumulate is a practical strategy.

Second, distributional drift should be monitored proactively. The strong MMD–AUROC correlation ($r = 0.90$, Table 7) enables a simple deployment strategy: periodically computing MMD between incoming frames and the training distribution, and triggering retraining when it exceeds a validation-calibrated threshold. Our temporal-bucket analysis indicates that monthly retraining captures most available performance, while quarterly retraining risks entering the steeper degradation regime.

Third, as Section VI demonstrates, no visual-only method can reliably detect semantic mismatch anomalies. Visual anomaly detection should therefore be paired with a separate order-verification module – a direction where VLM-anchored architectures (Section VII-C) appear particularly promising.

B. LIMITATIONS

RoboCook-TAD depicts a single menu item in a single robotic installation. While this scope is deliberate to isolate temporal drift (Section III-H), generalization to other dishes and deployment environments remains an open question that we intend to address through future multi-dish releases. The unified heatmap-to-bbox conversion pipeline also introduces a degree of approximation; method-specific conversions might improve some baselines, and full pipeline details are provided in the appendix. Additionally, the pixel-annotation feasibility study was conducted on only 80 instances, and a larger study might reveal sub-type-specific annotation strategies. Finally, all evaluations are frame-level; exploiting temporal consistency across consecutive frames in a video sequence is a natural extension.

C. FUTURE DIRECTIONS

The concave relationship between supervision level and temporal robustness suggests that neither end of the spectrum is optimal. Methods that continuously adapt to distributional drift, such as online memory-bank updates, test-time augmentation, or continual learning, may achieve a better trade-off, and RoboCook-TAD provides a ready-made testbed for evaluating such approaches. A second promising direction is VLM-anchored hybrid detection, combining a frozen VLM backbone for temporal stability with a periodically retrained task head for domain-specific accuracy; the Category C results further suggest that VLM features may detect semantic anomalies invisible to feature-based methods. Finally, as the benchmark expands to additional menu items, a key question is whether the temporal degradation mechanisms observed here (lighting drift, hardware aging, ingredient batch variation) transfer across dishes, in which case a single temporal evaluation protocol could serve an entire multi-dish benchmark family.

VIII. CONCLUSION

We have presented RoboCook-TAD, a 10,124-image benchmark from a commercially deployed udon-cooking robot that introduces temporal evaluation as the primary protocol for industrial anomaly detection. Our experiments yield three principal findings: (i) SOTA methods drop $\sim$21 AUROC points from MVTec AD, confirming that the field's near-saturation does not extend to deformable food; (ii) 2–10 points of apparent performance are artifacts of random splits, with degradation correlating strongly with measurable distributional drift ($r = 0.90$); and (iii) increased supervision does not monotonically improve temporal robustness – semi-supervised methods offer the best trade-off while fully-supervised detectors suffer the largest TGG. We release all data, annotations, and code, and hope this benchmark will encourage the community to evaluate anomaly detectors not only on static accuracy but on how gracefully they degrade as deployment conditions change.

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